HEALTH360: EXECUTIVE ANALYTICS & POPULATION HEALTH DASHBOARD

Final Project Report

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Team ID: 0827CY231062

Project Track: Data Analytics & Business Intelligence (Tableau Track)

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1. INTRODUCTION

1.1 Project Overview

Health360 is an enterprise-grade Business Intelligence solution engineered to transform static healthcare datasets into real-time, actionable population health insights. Built around a 5,000-patient synthetic dataset (smart_healthcare_dataset.csv), Health360 consolidates clinical vitals, lifestyle habit metrics, demographic classifications, and symptom prevalence into a single-screen Tableau Master Dashboard and a guided 9-point Executive Tableau Story.

1.2 Purpose

The purpose of Health360 is to bridge the gap between complex health data and strategic clinical decision-making. By providing dynamic global cross-filtering and pre-calculated composite risk logic, Health360 empowers executive leadership—specifically Chief Medical Officers, Wellness Directors, and Heads of Population Health—to identify high-risk patient cohorts, optimize resource allocation, and deploy targeted preventive healthcare interventions.

2. IDEATION PHASE

2.1 Problem Statement

Healthcare executives are constrained by fragmented, tabular data repositories that conceal critical multi-variable relationships. Static spreadsheets fail to isolate how behavioural factors (e.g., smoking and alcohol usage) compound with clinical metrics (e.g., Blood Pressure and Cholesterol) to elevate health risk scores across distinct demographic life stages (Children to Elder).

2.2 Empathy Map Canvas

- Who are we empathizing with? Dr. Michael Carter (CMO), Emily Rodriguez (Wellness Director), and Dr. Priya Nair (Head of Population Health).
- What do they need to do? Identify high-risk cohorts early, reallocate preventive care budgets efficiently, and reduce chronic condition readmissions.
- What do they see? Disjointed EHR tables, static monthly PDF reports, and rising chronic disease trends.
- What do they hear? Complaints about clinical resource shortages and demands for evidence-based wellness investments.
- What do they say and do? Conduct quarterly quality reviews, request ad-hoc data analysis, and present strategic proposals to health boards.
- Pains: Overwhelmed by tabular data volume; unable to perform real-time drill downs during executive meetings.

- Gains: Instant cohort filtering, visual risk stratification, and clear narrative storytelling.

2.3 Brainstorming

During the ideation phase, key visual metrics were prioritized:

1. Clinical Layer: Scatter plot mapping Blood Pressure against Cholesterol with color-coded composite risk scores.
2. Behavioural Layer: Matrix heatmaps and stacked bar charts evaluating substance co-occurrence (Smoking vs. Alcohol) and activity levels.
3. Demographic Layer: Age/gender distribution pyramids and BMI pie charts.
4. Symptom Layer: Multi-metric bar charts tracking acute symptoms (Chest Pain, Dizziness, Fatigue) across life stages.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The user journey maps the end-to-end executive interaction:

- **Awareness:** Executive notices rising chronic disease rates and needs cohort visibility.
- **Ingestion:** Data is loaded via high-performance .hyper extracts.
- **Exploration:** Executive uses KPI cards and cross-filters the Master Dashboard to isolate risk drivers.
- **Action:** Executive navigates the 9-point Tableau Story to present findings to board members.

3.2 Solution Requirements

- **Functional Requirements (FR):** Data ingestion (5,000 records), calculated measure creation (Age Category, Calculated Risk Score), 9 visual worksheets, dynamic global cross-filtering, and cloud deployment on Tableau Public.
- **Non-Functional Requirements (NFR):** High usability (<1 sec filter latency), anonymized security compliance (no PII), 24/7 cloud availability, and scalability up to 100,000+ records.

3.3 Data Flow Diagram (DFD)

- **DFD Level 0:**
  - **DS-1** → smart_healthcare_dataset.csv → **Process 1.0 (Data Ingestion & Transformation)** → **DS-2:** Tableau Extract (.hyper).

- **DS-2 → Process 2.0 (Visual Aggregations) → Process 3.0 (Master Dashboard & Story Engine) → Executive Personas**.

3.4 Technology Stack

- **Query & Business Logic:** Tableau Calculated Fields Syntax.
- **Data Layer:** Flat-file CSV (smart_healthcare_dataset.csv) & Tableau Data Extract (.hyper).
- **Visualization Engine:** Tableau Desktop Professional.
- **Deployment Platform:** Tableau Public Cloud Engine.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Health360 addresses executive pain points by converting flat data into an interactive single-screen dashboard and narrative story, enabling single-click cohort segmentation and risk profiling without requiring custom SQL or coding.

4.2 Proposed Solution

An end-to-end Business Intelligence workflow featuring:

1. Pre-calculated metrics: Age Category (Children to Elder) and Calculated Risk Score.
2. 9 dedicated visual worksheets spanning vitals, habits, demographics, and symptoms.
3. Integrated single-page Master Dashboard with full cross-filtering.
4. Guided 9-point Executive Tableau Story.

4.3 Solution Architecture

[DATA SOURCE] → smart_healthcare_dataset.csv (5,000 Patient Records)

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▼

[INGESTION] → Tableau Desktop Native CSV Connector

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▼

[PREPARATION] → Calculated Fields Syntax (Age Category, Calculated Risk Score)

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▼

[STORAGE] → Tableau Data Extract (.hyper Columnar Engine)

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▼

[BI & ANALYTICS] → 9 Worksheets + Master Dashboard + 9-Point Executive Story

|

▼

[ENDPOINTS] → Tableau Public Cloud Server (Browser / Tablet Access)

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Structured across 4 agile sprints completed as a solo contributor:

- **Sprint 1 (Days 1–5):** Data ingestion, validation, calculated field setup (Age Category, Risk Score), and .hyper extract creation.
- **Sprint 2 (Days 6–10):** Visual development of all 9 dedicated worksheets (vitals scatter plot, lifestyle heatmap, disease prevalence, demographic pyramid, symptom analysis).
- **Sprint 3 (Days 11–15):** Assembly of single-screen Master Dashboard and implementation of dynamic global action filters.
- **Sprint 4 (Days 16–20):** Composition of the 9-point Executive Tableau Story, cloud deployment to Tableau Public, and performance testing.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

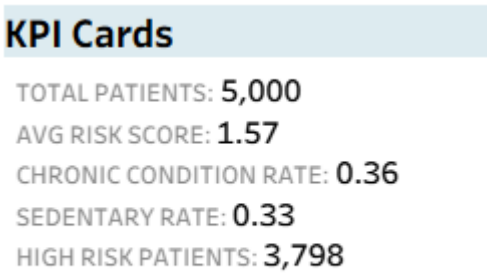
- **Data Volume Rendered:** 5,000 synthetic patient records.
- **Initial Board Load Time:** < 1.8 seconds.
- **Extract Query Latency:** < 120 ms using Tableau .hyper engine.
- **Cross-Filter Response Time:** < 0.5 seconds per user interaction.
- **Calculations Verified:** Age Category, Calculated Risk Score, Chronic Condition Rate, Sedentary Rate.
- **Visual Components Tested:** 9 Worksheets, 1 Master Dashboard, 9 Story Points.

7. RESULTS

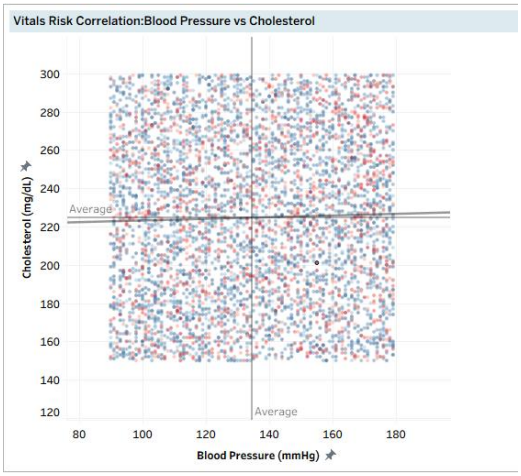
7.1 Output Screenshots

(Include screenshots corresponding to each section below)

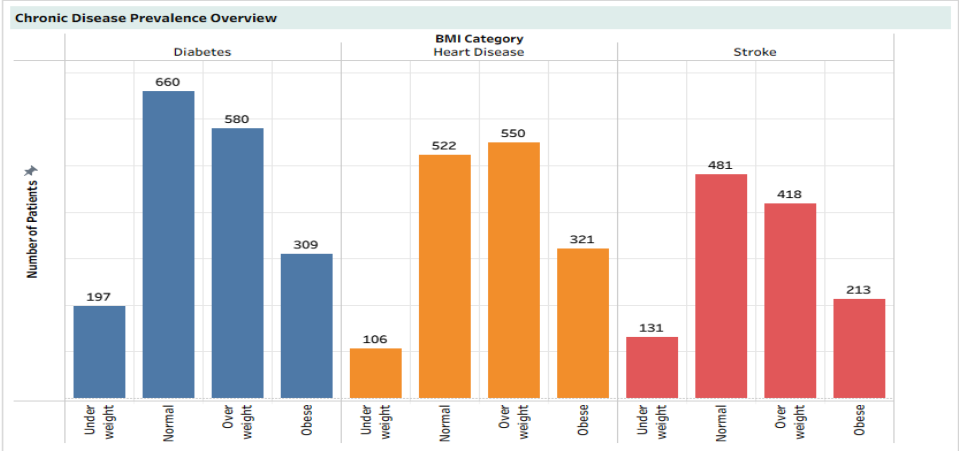
1. **Executive KPI Cards:** Scorecards displaying Total Patients (5,000), Avg Risk Score (1.57), Chronic Condition Rate (0.36), Sedentary Rate (0.33), and High Risk8 Patients (3,798).



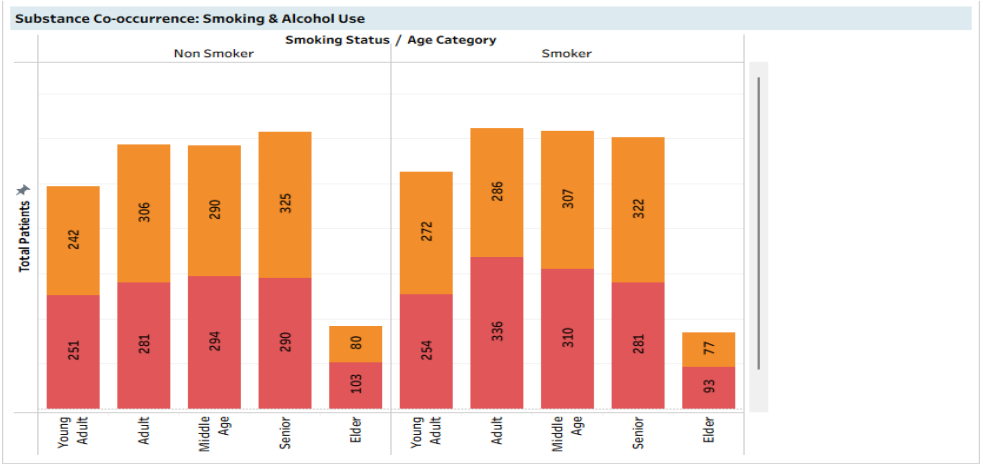
2. **Vitals Risk Correlation:** Scatter plot mapping Blood Pressure vs. Cholesterol color-coded by risk.



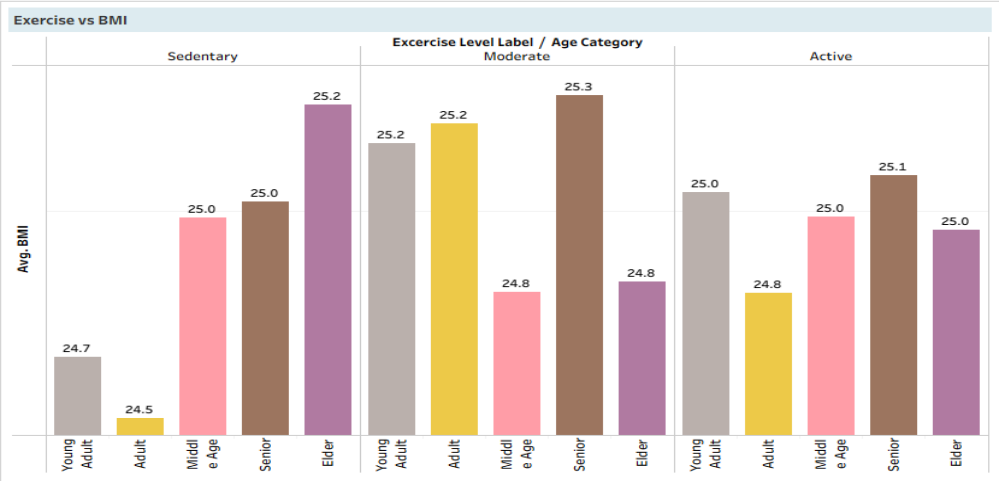
3. **Disease Prevalence Overview:** Grouped bar chart tracking Diabetes, Heart Disease, and Stroke across BMI categories.



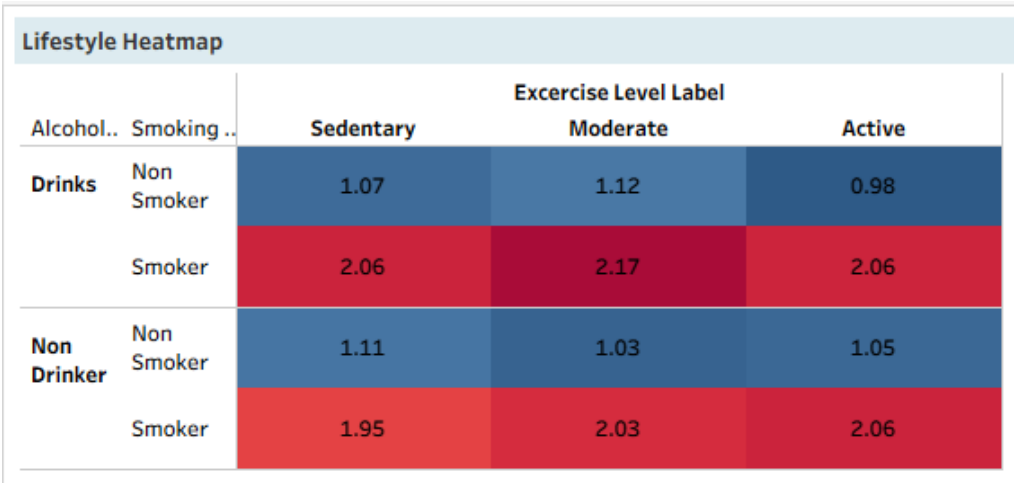
4. **Substance Co-occurrence:** Stacked bar chart analysing Smoking vs. Alcohol across age brackets.



5. **Exercise vs. BMI:** Bar chart comparing average BMI across exercise levels and age groups.



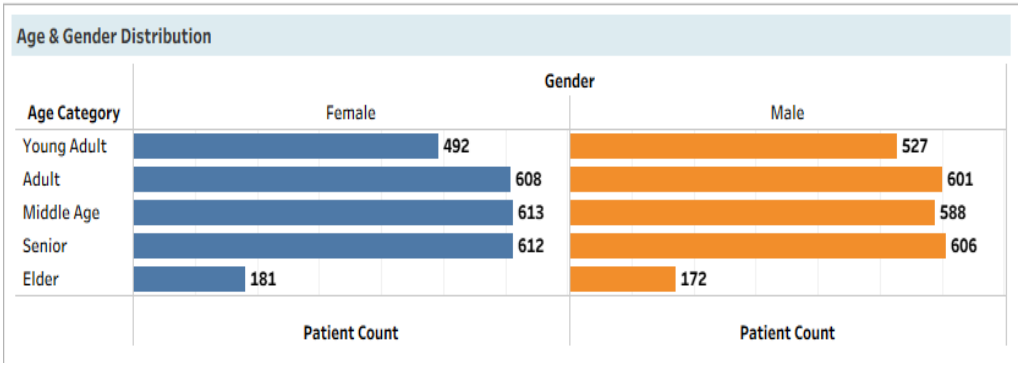
6. **Lifestyle Heatmap:** 2-way matrix cross-tabulating habits against average risk score.



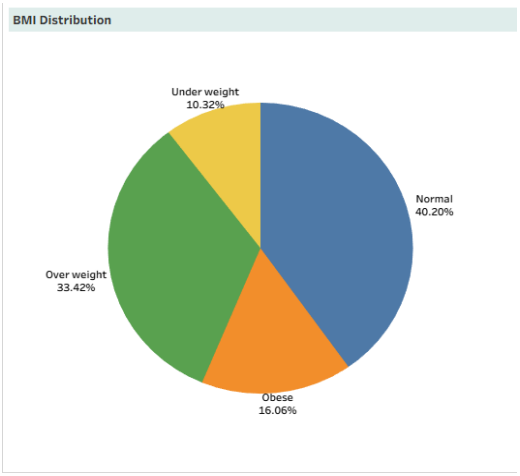
Lifestyle Heatmap

Alcohol..	Smoking ..	Excercise Level Label: Sedentary	Moderate	Active
Drinks	Non Smoker	1.07	1.12	0.98
	Smoker	2.06	2.17	2.06
Non Drinker	Non Smoker	1.11	1.03	1.05
	Smoker	1.95	2.03	2.06

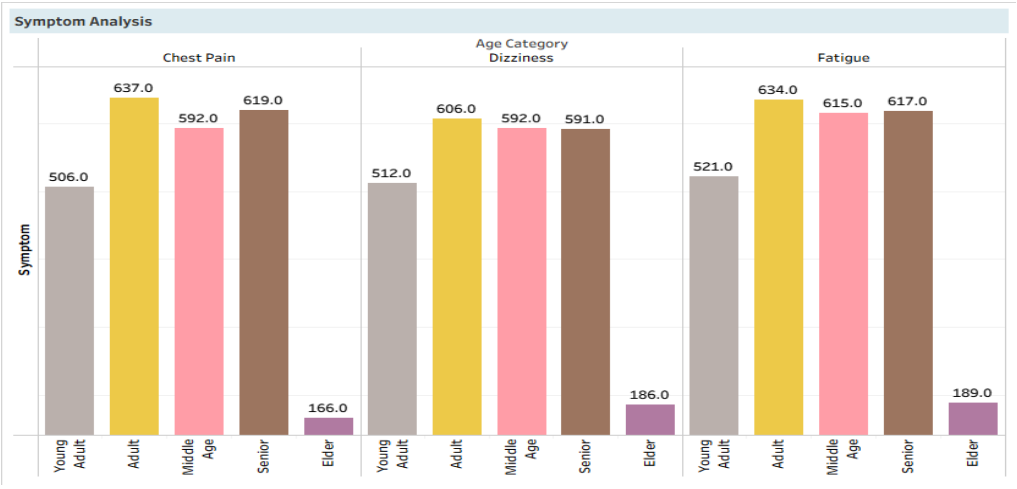
7. **Age & Gender Distribution:** Diverging population pyramid.



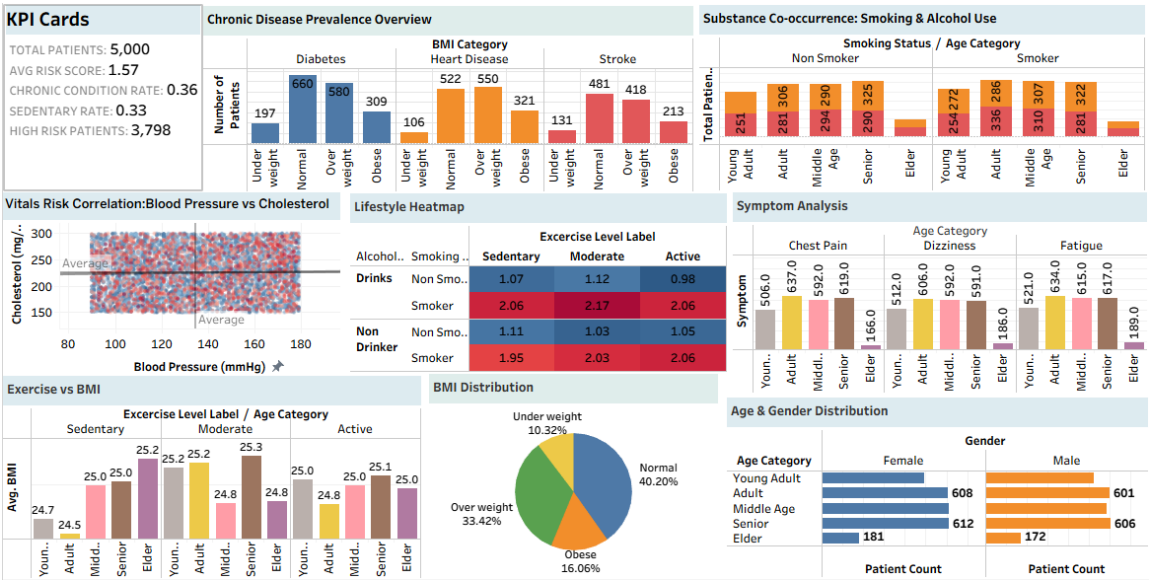
8. **BMI Distribution:** Pie chart breaking down weight classifications (Normal 40.20%, Overweight 33.42%, Obese 16.06%, Underweight 10.32%).



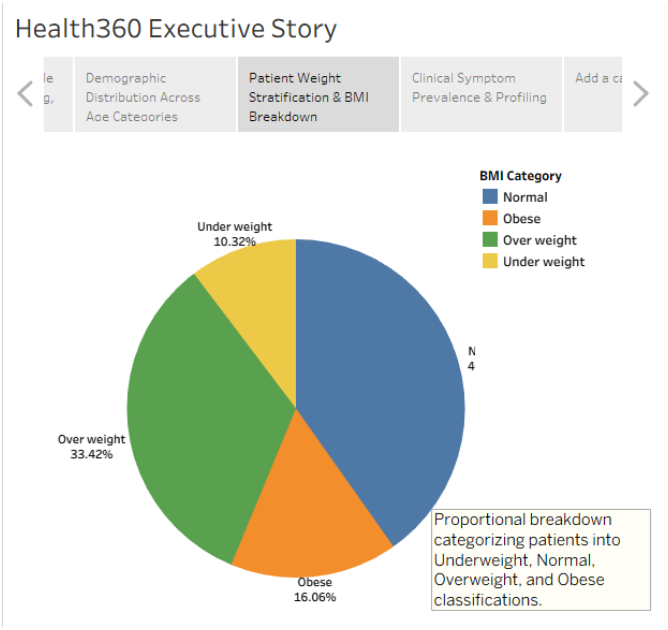
9. **Symptom Analysis:** Bar chart displaying Chest Pain, Dizziness, and Fatigue across life stages.



10. **Health360 Master Dashboard:** Single-screen integrated executive interface.



11. **Health360 Executive Story:** 9-point interactive storyboard.



8. ADVANTAGES & DISADVANTAGES

Advantages

- **Unified Single-Screen View:** Eliminates multi-tab clutter by consolidating 9 analytical domains into one view.
- **Sub-Second Interactivity:** Pre-computed .hyper extracts ensure instant cross-filtering response times.
- **Actionable Risk Profiling:** Multi-factor calculated risk scoring provides immediate triage insights.
- **Zero-Footprint Deployment:** Accessible via any standard web browser through Tableau Public Server.

Disadvantages

- **Static Flat-File Ingestion:** Current build relies on manual batch updates of the CSV file rather than an automated live database feed.
- **Synthetic Cohort Limits:** Synthetic dataset distributions may not fully capture edge-case clinical anomalies found in real-world EHRs.

9. CONCLUSION

The **Health360 Executive Analytics & Population Health Dashboard** successfully solves the challenge of converting complex, multi-dimensional patient data into actionable clinical insights. By integrating pre-calculated composite risk scoring, dynamic cross-filtering, and a 9-point narrative story, Health360 equips healthcare

decision-makers with the tools required to identify risk factors early, optimize resource management, and improve patient health outcomes.

10. FUTURE SCOPE

1. **Live Database Integration:** Connect the Tableau pipeline directly to enterprise data platforms (e.g., Snowflake, PostgreSQL, or AWS Redshift) for real-time EHR streaming.
2. **Predictive Machine Learning Integration:** Incorporate predictive risk models (e.g., Python/R script integration via TabPy) to forecast 5-year patient chronic disease onset probabilities.
3. **Automated Alerting Engine:** Implement automated email/SMS notifications when patient risk scores breach defined clinical thresholds.

11. APPENDIX

Dataset

- **Dataset Name:** smart_healthcare_dataset.csv (5,000 anonymized patient records)

Project Demo Links

- **Tableau Public Dashboard:** https://public.tableau.com/app/profile/sarthak.siddharth.sahare/viz/Health360-ComprehensivePatientAnalyticsDashboard/Health360MasterDashboard
- **Tableau Public Story:** https://public.tableau.com/app/profile/sarthak.siddharth.sahare/viz/Health360PatientInsightsRiskStory/Health360ExecutiveStory